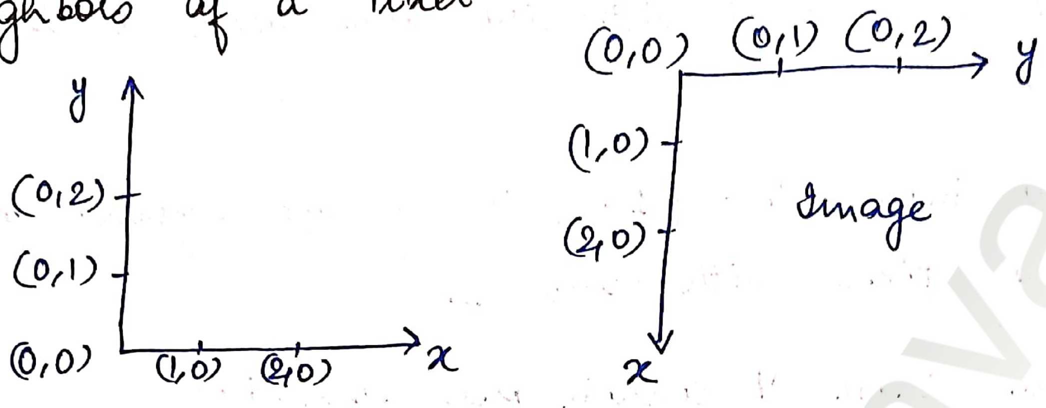
Some Basic Relationship between pixel

→ Neighbors of a Pixel



- A pixel p at coordinate (x,y) has two horizontal and two vertical Neighbors with co-ordinates

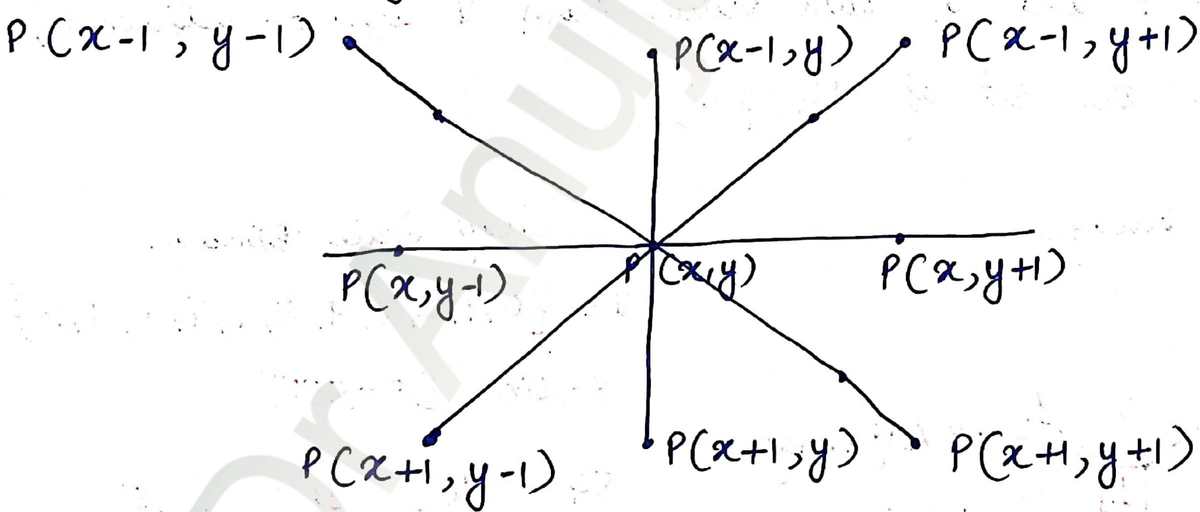
$$(x+1, y),\ (x-1, y),\ (x, y+1),\ (x, y-1)$$

This set of pixels, called 4 Neighbors of p, $N_4(p)$.

- The four diagonal neighbors of p have coordinates

$$(x+1, y+1),\ (x+1, y-1),\ (x-1, y+1),\ (x-1, y-1)$$

are denoted by $N_D(p)$



- 8 Neighbors of p ⇒ $N_8(p) = N_4(p) + N_D(p)$

Adjacency

- Let V be the set of intensity values used to define adjacency.
- For binary image, $V = \{1\}$, if we are referring to adjancey of pixels with value 1.
- For grayscale image, V contain more elements.

  eg for adjancey of pixel in the range 0 to 255, set V could be any subset of these 256 values.

- Three - Types of Adjacency

(i) 4 - Adjacency

Two pixels p and q with values from V are 4 - adjacent if q in the set of $N_4(p)$.

(ii) 8 - Adjacency

Two pixels p and q with values from V are 8 - adjacent, if q in the set of $N_8(p)$.

eg Binary Image

$V = \{1\}$

0	1	0	1
0	0	$\boxed{1}$	0
0	0	1	0
1	0	0	0

8 Adjacency

4 Adjacency

Not Connected.

Gray Scale Image

$V = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

8-bit [0-255]

54	10	100	5
81	150	$\boxed{2}$	34
201	200	3	45
7	70	147	56

8 Adj.

4 Adj.

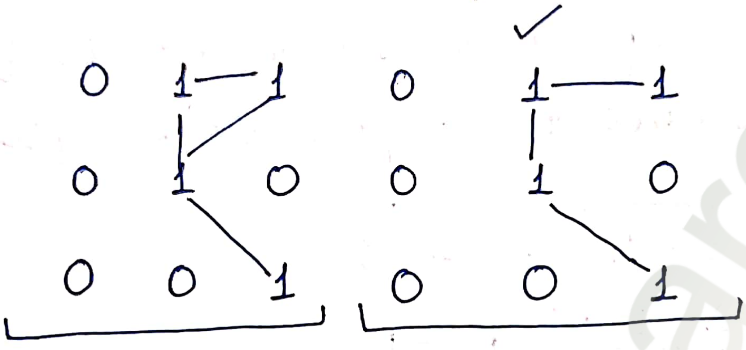
Not Connected

(iii) m- Adjacency (Mixed Adjacency)

Two Pixels p and q with values from V are m-adjacent if

(a) q is in $N_4(p)$ or

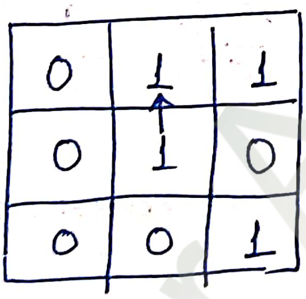
(b) q is in $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixels whose values are from V.



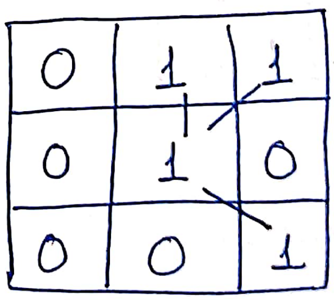
eg Find 4- adjacency & 8 adjacency of the center pixel. for V={1}

0	1	1
0	1	0
0	0	1

Soln:



4 Adjacency.



8 - Adjacency

Connectivity

- Two pixels p & q with coordinates (x_0, y_0) & (x_n, y_n) are said to be connected if there exist a path.

Path

- A path from pixel p to pixel q is a sequence of distinct pixels.

eg Consider the image segment as shown. Let $V = \{0, 1\}$. Compute the length of the shortest 4 - 8 path between pixel p and q.

- If a particular path does not exist b/w these two points, explain why.

3	1	2	1	q
2	2	0	2	
1	2	1	1	
1	0	1	2	

p

Soln:

4 - Path

3	1	2	1	q
2	2	0	2	
1	2	1	1	
1	0	1	2	

p

Not Possible

8 - Path

3	1	2	1	q
2	2	0	2	
1	2	1	1	
1	0	1	2	

p

✓ Length → 4

→ When $V = \{0, 1\}$, 4 path does not exist between p and q because it is impossible to get from p and q by travelling along points that are both 4- adjacent and have values from V. From fig, it is not possible to get to q.

→ The shortest 8- path is 4.

for m - path

3	1	2	1
2	2	0	2
1	2	1	1
1	0	1	2

The length of the shortest m-path is 5.

Repeat above example for $V = \{1, 2\}$

4 - Path

	3	1	2	1 q
	2	2	0	2
	1	2	1	1
p	1	0	1	2

Length - 6

8 - Path

	3	1	2	1 q
	2	2	0	2
	1	2	1	1
p	1	0	1	2

Length - 4

M - Path

	3	1	2	1 q
	2	2	0	2
	1	2	1	1
p	1	0	1	2

Length - 5

eg Consider two image subsets S1 and S2 and the figure. For $V = \{1\}$ determine whether these two subsets are

a) 4 - Adjacent b) 8 - Adjacent c) m - Adjacent

	S_1				S_2				
0	0	0	0	0	0	0	1	1	0
1	0	0	1	0	0	1	0	0	1
1	0	0	1	0	1	1	0	0	0
0	0	1	1	1	0	0	0	0	0
0	0	1	1	1	0	0	1	1	1

Soln: Let p & q be as shown in fig., then.

a) S_1 & S_2 are not 4 - connected because q is not in set $N_4(p)$

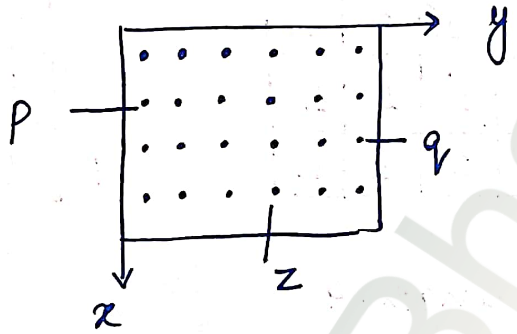
b) S_1 & S_2 are 8 - connected because q is in the set $N_8(p)$.

c) S_1 & S_2 are m - connected because (i) q is in $N_D(p)$ and (ii) $N_4(p) \cap N_4(q)$ is empty.

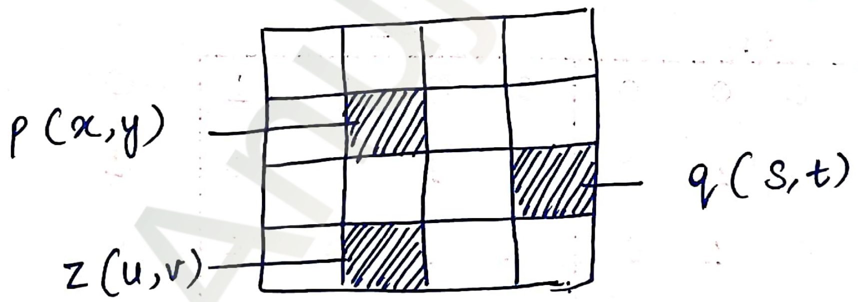
			S_1				S_2		
0	0	0	0	0	0	0	1	1	0
1	0	0	1	0	0	1	0	0	1
1	0	0	1	0	1 q	1	0	0	0
0	0	1	1	p 1	0	0	0	0	0
0	0	1	1	1	0	0	1	1	1

Distance Measure

- Given, an image $f(x,y)$ graphically represented as



Consider pixels p, q, z with coordinates (x,y), (s,t) & (u,v) then the distance function D has the following properties :



(i) $D(p,q) \geq 0$ $\quad [\,D(p,q) = 0$, iff $p = q\,]$

(ii) $D(p,q) = D(q,p)$

(iii) $D(p,z) \leq [\,D(p,q) + D(q,z)\,]$

The following are the different distance measures.

1. Euclidean Distance

$$D_E(p,q) = [(x-s)^2 + (y-t)^2]^{1/2}$$

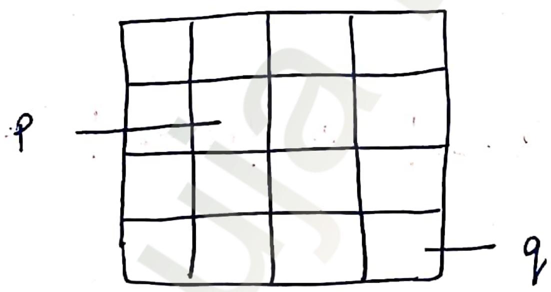
2. City Block Distance

$$D_4(p,q) = |x-s| + |y-t|$$

3. Chess Board Distance

$$D_8(p,q) = \max(|x-s|, |y-t|)$$

eg Consider an image as shown. Calculate the distance measure : (i) Euclidean Distance (ii) City Block distance (iii) Chess Board.



Soln:- From above fig. $p(2,2)$ & $q(4,4)$

(i) Euclidean distance

$$D_E(p,q) = [(2-4)^2 + (2-4)^2]^{1/2}$$

$$= \sqrt{8}$$

ii) City Block Distance

$$D_4(p,q) = |2-4| + |2-4|$$

$$= 4$$

(iii) Chess Board Distance

$$D_8(p,q) = \max(|2-4|, |2-4|)$$
$$= 2$$

eg Consider the following image & find the distance between p and q.

$$\begin{array}{cccc} p & \begin{array}{|c|c|c|}\hline 0 & 0 & 1 \\ \hline 1 & 1 & 0 \\ \hline 1 & 1 & 0 \\ \hline \end{array} & q \end{array}$$

Soln:- $p(0,0)$ & $q(2,2)$

(i) Euclidean distance

$$D_E(p,q) = \left[(0-2)^2 + (0-2)^2\right]^{1/2} = \sqrt{8}$$

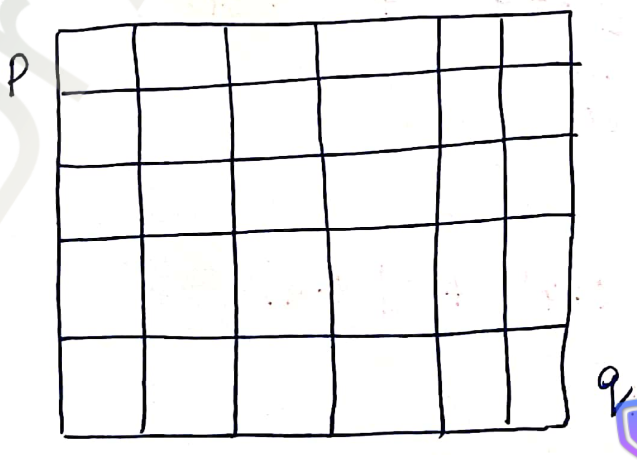
(ii) City - Block Distance

$$D_4(p,q) = |(0-2)| + |(0-2)| = 4$$

(iii) Chess Board distance

$$D_8(p,q) = \max\left[|0-2|, |0-2|\right] = 2$$

eg Consider the image & find distance between p and q.



Soln: from above fig., $p(0,0)$ & $q(4,5)$

i) Euclidean distance

$$D_E(p,q) = \left[(0-4)^2 + (0-5)^2\right]^{1/2}$$

$$= \sqrt{41}$$

ii) City Block distance

$$D_4(p,q) = |0-4| + |0-5|$$

$$= 9$$

iii) Chess Board distance

$$D_8(p,q) = \max(|0-4|, |0-5|)$$

$$= 5$$

Representation

(1) City Block Distance

4		2		4
	2	1	2	
2	1	0	1	2
	2	1	2	
4		2		4

(ii) Chess Board Distance

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

eg Consider the image segment shown below

a) let $V = \{0, 1\}$ and Compute D_4, D_8 and D_m distance between p and q.

b) Repeat for $V = \{1, 2\}$

$$
\begin{array}{ccccc}
 & 3 & 1 & 2 & \boxed{1} \; q \\
 & 2 & 2 & 0 & 2 \\
 & 1 & 2 & 1 & 1 \\
p & \boxed{1} & 0 & 1 & 2
\end{array}
$$

Soln: $p \rightarrow (3, 0)$ $\quad q\ (0, 3)$

$D_4(p, q) = |0-3| + |0-3| = 6$

$D_8(p, q) = \text{Max}(|0-3|, |0-3|) = 3$

$D_m(p, q) = 5$ $\quad \because$

$$
\begin{array}{ccccc}
 & 3 & 1 & 2 & 1 \; q \\
 & 2 & 2 & 0 & 2 \\
 & 1 & 2 & 1 & 1 \\
p & 1 & 0 & 1 & 2
\end{array}
$$

b) D_4 and D_8 are independent of V, so they are same as above

for D_m

	3	1	2	1	q
	2	2	0	2	
	1	2	1	1	
p	1	0	1	2	

$D_m = 6$